

Model Selection Memo

Recommendation → I recommend XGBoost over Random Forest model, because XGBoost achieved a better test ROC-AUC while maintaining an acceptable train-test gap, indicating stronger predictive performance on unseen data. ROC-AUC is the main metric I used to compare how well the models distinguished between passengers who survived and those who did not.

Model Performance →

<u>Model</u>	<u>Test ROC-AUC</u>	<u>Train ROC-AUC</u>	<u>Train-Test Gap</u>
Random Forest	0.740	1.000	0.8311
XGBoost	0.735	0.9518	0.8451

The XGBoost model was selected because its test performance was similar compared with Random Forest, while the train-test gap was acceptable.

Recommended Model Configuration →

The recommended XGBoost model uses the following hyperparameters :-

- ⑩ Model → XGBClassifier
- ⑩ n_estimators → 1000
- ⑩ learning_rate → 0.1
- ⑩ max_depth → 3
- ⑩ eval_metrics → "logloss"
- ⑩ random_state → 42
- ⑩ early_stopping_rounds → 30

The tree count was not manually fixed at 1000. Instead, 1000 was used as an upper limit and early stopping selected the best iteration based on validation performance. The model stopped after the validation score stopped improving for 30 consecutive rounds.

Main Risk → One risk is that XGBoost can overfit, especially with too many trees or overly complex trees. Although early stopping helps control this risk, performance should still be monitored on new production data. Boosting can become too closely fitted to the training data when the model is too deep or trained for too many rounds.

Next Test → If I had another day, I would test the model on a separate validation set and perform controlled hyperparameter tuning, especially for learning_rate, max_depth, subsample and the number of estimators. This would help determine whether the current performance can be improved without increasing overfitting.

Feature Findings → The feature importance and permutation-importance analysis showed that sex_male was the strongest feature. Both Random Forest and XGBoost identified important relationships involving the same major Titanic features, although their exact rankings were not completely identical. The partial dependence plots for sex_male were also very similar. Both models learned that Female passengers (sex_male = 0) has a substantially higher predicted probability of survival than male passengers (sex_male = 1).

Final Decision → I would put XGBoost model into production because it achieved the stronger test ROC-AUC by 0.735, while maintaining a train-test gap of 0.8451. However,

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this recommendation should be validated on additional data before deployment because the current dataset is small and the evaluation uses a single train/test split.

Based on predictive performance, generalization, and the results of the feature-importance analysis, XGBoost is my recommended model for production. I would continue monitoring its performance after deployment and reassess the model if its performance changes on new data.